Overweightness and SES

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Contents

```r
require(tidyverse)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```r
require(yaml)
```

```
## Loading required package: yaml
```

```r
library(readxl)
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```r
library(patchwork)
#tinytex::install_tinytex()
```

Overweightness and SES

1 Overweightness in the Netherlands: A Social Problem Hidden in Plain Sight

Walk through any Dutch city today and the numbers get to ignore. More than half of Dutch adults are overweight, one in six has obesity (Rijksinstituut voor Volksgezondheid en Milieu [RIVM], 2026), and if current trends continue, two in three people could be affected by 2050. What was once seen as a personal health issue has become a major social challenge. This became clear when Zorginstituut Nederland announced that bariatric surgery will now be covered by basic health insurance for teenagers aged 13 to 18 with severe obesity. Although only a small number of young people will qualify, the decision reflects growing recognition that lifestyle interventions alone are not always enough. To understand obesity, however, we must look beyond individual choices. In the Netherlands, obesity is strongly linked to social inequality. People with lower levels of education are significantly more likely to be overweight than university graduates (RIVM, 2026). Lower-income groups often have less access to healthy food, safe places to exercise, and supportive living environments. Research also shows that factors such as neighborhood safety, green spaces, and social cohesion influence obesity rates.(OECD & European Observatory on Health Systems and Policies, 2025). The consequences extend beyond health. Obesity is associated with higher healthcare costs, lower earnings, increased sick leave, and reduced economic productivity. Children with obesity are more likely to face educational, social, and health disadvantages later in life. This argues that obesity in the Netherlands should be viewed as a social problem rather than simply a matter of personal responsibility. Addressing it effectively requires tackling the structural inequalities that make obesity more common among disadvantaged groups.

2 Data

2.1 Loading in data.

```
gewicht_regios <-
  read_xlsx("data/Gezondheidsmonitor__regio__2024_04062026_114221.xlsx")
saveRDS(gewicht_regios, "data/gewicht_regios.rds")

overgewicht_over_tijd <-
  read_xlsx("data/Lengte__onder__en_overgewicht_vanaf_1981_04062026_120636.xlsx")
saveRDS(overgewicht_over_tijd, "data/overgewicht_over_tijd.rds")

inkomen_dataset <-
  read_xlsx("data/Inkomens_Huishouden.xlsx")
saveRDS(inkomen_dataset, "data/inkomen_dataset.rds")

opleiding_dataset <-
  read_xlsx("data/Onderwijs_per_provincie.xlsx")
saveRDS(opleiding_dataset, "data/opleiding_dataset.rds")

gewicht_regios <-
  readRDS("data/gewicht_regios.rds")
overgewicht_over_tijd <-
  readRDS("data/overgewicht_over_tijd.rds")
inkomen_dataset <-
  readRDS("data/inkomen_dataset.rds")
opleiding_dataset <-
  readRDS("data/opleiding_dataset.rds")
```

loading in the data and saving it as an RDS so that they will not be lost

2.2 Explanation data

In this project, we utilize two datasets released by CBS (Statistics Netherlands) that collectively offer insight into the health of the Dutch population. The initial dataset, Gezondheidsmonitor; adults 18 years and older, region, 2024, centers on individuals aged 18 and above residing in the Netherlands. CBS compiled it in partnership with the National Institute for Public Health and the Environment (RIVM) and the local Public Health Services (GGDs). The data collection provides insights into health conditions, lifestyle choices, and social elements across various geographical tiers, such as municipalities, GGD regions, provinces, and at the national level. Essential indicators consist of self-assessed health, rates of chronic diseases, overweight and obesity, levels of physical exercise, involvement in sports, tobacco use, alcohol intake, and informal caregiving. The information is primarily shown as percentages of the adult demographic and offers a comprehensive regional overview of health and wellness in 2024.The second dataset, Lengte en gewicht van personen, ondergewicht en overgewicht; vanaf 1981, provides a longitudinal view of body weight and obesity trends in the Netherlands. This dataset includes individuals aged four and older and monitors trends from 1981 to 2025. It contains data regarding body mass index (BMI), moderate overweight, obesity, and underweight, categorized by age group, gender, and year. The data relies on self-reported weight and height, which CBS utilizes to determine BMI categories. In contrast to the initial dataset that highlights regional variations for one year, this dataset offers a national time series enabling researchers to explore changes in overweight and obesity across more than forty years.

Collectively, these datasets enhance each other effectively. The Health Monitor dataset offers an in-depth view of regional health disparities in 2024, whereas the height and weight dataset uncovers enduring national patterns of overweight and obesity. Integrating information from both sources allows for a clearer comprehension of obesity's prevalence in the Netherlands, its regional variations, and its historical trends.

3 Quantifying

3.1 Data cleaning and creating new variables

```
overgewicht_1981 = overgewicht_over_tijd[1,5]
overgewicht_2025 = overgewicht_over_tijd[13,5]
```

Selecting data for making the variable for the percentage change over time (later in the PDF)

```
perc_overgewicht <-
  full_join(overgewicht_over_tijd[1:13, c(4,5,7)],
            full_join(overgewicht_over_tijd[40:52, c(4,5)],
                      overgewicht_over_tijd[79:91, c(4,5)],
                      by = "Perioden"), by = "Perioden")
```

Full joining by "Perioden" to make one data set of all the necessary data with the same left column, the year it was in. The data is the year, percentage of total overweight people in the Netherlands, percentage of obesity in the Netherlands, percentage of overweight men and lastly percentage of overweight women in the Netherlands.

```
list_perc_point_change <-  numeric(12)
for (i in 1:12) {
  list_perc_point_change[i] <- perc_overgewicht[[2]][i + 1] - perc_overgewicht[[2]][i]
}

perc_overgewicht$Percentage_difference <- c(NA, list_perc_point_change)
```

To the dataset the percentage point change of overweight people compared to the last datapoint is added. Since there is no percentage point change for the first datapoint, NA gets added there. This will be used for the last plot. This is the first final dataset.

```
opleiding_provincie <-
  data.frame("Regio" = opleiding_dataset[[4]][6:17],
             "Percentage_laagopgeleide" = opleiding_dataset[[6]][6:17] +
              opleiding_dataset[[7]][6:17],
             "Percentage_middelopgeleide" = opleiding_dataset[[8]][6:17],

             "Percentage_hoogopgeleide" = opleiding_dataset[[9]][6:17] +
              opleiding_dataset[[10]][6:17]
)
```

The education levels dataset of the Netherlands consists of 6 different columns that are used. First, completely on the left, the provences, because this is what will be used later, To get the data we want to use 2 new variables need to be created, firstly the sixth and seventh column need to be added together, to create the percentage of 'lower educated' people in the Netherlands, and then the ninth and tenth column need to be added to get the percentage of 'higher educated' people in the Netherlands. This get combined into one dataset with data.frame with the percentage of 'middle educated' people in the Netherlands. Only rows 6 till 17 are used, because this is the data of the different provences in the Netherlands.

```
gemiddeld_inkomen <- data.frame(Regio = inkomen_dataset[[4]][2:13],
                                "Gemiddeld_inkomen(x 1000)" = inkomen_dataset[[6]][2:13])
```

The income dataset is used in a similar way as the education levels dataset. A new data set gets created with data.frame, on the left again the provences, called "Regio", or region in english. Next to that the average income per person of said provence divided by 1000 is there.

```
data_regios <- gewicht_regios[grepl("\\(GG\\)", gewicht_regios[[3]] ), 3:12]
```

the last dataset that was loaded in is divided in GGD regions instead of provences, because that is how the data is collected. In this dataset there is also differerent data of other regions which are not needed. The GGD data that is needed has GG in the names. the line "gewicht_regios[grepl("\(GG\)", gewicht_regios[[3]]), 3:12]" selects that data so that it can be used in the coming chunks.

The dataset (data_regios) is built up in regions instead of provences. To compare this dataset to the other datasets, which are built up in provences, the regions need to be transformed into provences aswell. This is done with the following data from AlleCijfers.nl (AlleCijfers, 2026).

```
groningen <- 601286

friesland <- 665208

drenthe <- 508109

ijsselland <- 555879
twente <- 646496

flevoland <- 462816

noord_oost_gelderland <- 851876
gelderland_midden <- 731773
gelderland_zuid <- 589699

utrecht <- 1415104

hollands_noorden <- 687079
zaanstreek <- 354021
kennemerland <- 570752
amsterdam <- 1151282
gooi <- 246252

haaglanden <- 1176111
hollands_midden <- 854306
rotterdam <- 1376801
zuid_holland_zuid <- 475490

zeeland <- 394538

west_brabant <-734825
hart_brabant <- 1122960
```

```r
brabant_zuidoost <- 819169

limburg_noord <- 537019
limburg_zuid <- 601355
```

```r
data_analysis <- function(regio1 = 0, data1 = 0, regio2 = 0, data2 = 0,
                          regio3 = 0, data3 = 0, regio4 = 0, data4 = 0,
                          regio5 = 0,  data5 = 0){
  totaal_provincie <- regio1 + regio2 + regio3 + regio4
  data_provincie <- ((regio1 / totaal_provincie) * data1)
  + ((regio2 / totaal_provincie) * data2)
  + ((regio3 / totaal_provincie) * data3)
  + ((regio4 / totaal_provincie) * data4)
  + ((regio5 / totaal_provincie) * data5)

  return(data_provincie)

}
```

The function above is created to calculate a new variable, the percentage of overweight people in the provinces instead of the percentage of overweight people in GGD regions. These GGD regions are always in one province, making it fairly easy to calculate the percentage of the provinces. These provinces are made up of different amount of GGD regions, so the input data is standardized to zero for everything, so that one to five different factors can be given without the function giving an error. This then returns the percentage of overweight people in the province by the formula in the chunk.

```r
gezondheids_data_provincies <-
  rbind(Groningen  = data_analysis(groningen, data_regios[1, 2:10]),

        Fryslân = data_analysis(friesland, data_regios[24, 2:10]),

        Drenthe = data_analysis(drenthe , data_regios[2, 2:10]),

        Overijssel = data_analysis(ijsselland, data_regios[3,2:10],
                                   twente, data_regios[4,2:10]),

        Flevoland = data_analysis(flevoland , data_regios[8, 2:10]),

        Gelderland = data_analysis(noord_oost_gelderland, data_regios[5, 2:10],
                                   gelderland_midden, data_regios[6, 2:10],
                                   gelderland_zuid, data_regios[7, 2:10]),

        Utrecht = data_analysis(utrecht, data_regios[9, 2:10]),

        "Noord-Holland" = data_analysis(hollands_noorden , data_regios[10, 2:10],
                                        zaanstreek, data_regios[25, 2:10],
                                        kennemerland, data_regios[11, 2:10],
                                        amsterdam, data_regios[12, 2:10],
                                        gooi, data_regios[13, 2:10]),

        "Zuid-Holland" = data_analysis(haaglanden, data_regios[23, 2:10],
                                       hollands_midden, data_regios[14, 2:10],
                                       rotterdam, data_regios[15, 2:10],
```

```
                                    zuid_holland_zuid, data_regios[16, 2:10]),

        Zeeland = data_analysis(zeeland , data_regios[17, 2:10]),

        "Noord-Brabant" = data_analysis(west_brabant, data_regios[18, 2:10],
                                        hart_brabant, data_regios[19, 2:10],
                                        brabant_zuidoost, data_regios[20, 2:10]),

         Limburg = data_analysis(limburg_noord, data_regios[21, 2:10],
                                 limburg_zuid, data_regios[22, 2:10]))

gezondheids_data_provincies <- as.data.frame(gezondheids_data_provincies)

gezondheids_data_provincies$Provincie <- rownames(gezondheids_data_provincies)
rownames(gezondheids_data_provincies) <- NULL
gezondheids_data_provincies <-
  gezondheids_data_provincies[, c("Provincie",
  setdiff(names(gezondheids_data_provincies), "Provincie"))]
```

A new dataset gets created, with the function from before. Every name is in a very specific way to merge it into a bigger dataset for the different maps that need to be created. This is also the explanation for line 200 to 205, to merge with columnname "provincie".

```
data_overweight_provence <- data.frame(Regio = gezondheids_data_provincies[[1]],
  Overgewicht = gezondheids_data_provincies[[4]] + gezondheids_data_provincies[[5]])
```

The complete dataset that was created above this, is cut down to only the necessary information, the percentage of overweight people, the sum of column 4 and 5.

```
name_change <- full_join(opleiding_provincie, gemiddeld_inkomen)
```

```
## Joining with `by = join_by(Regio)`
```

The dataset of education level and average income do not have the right name yet for their left column to merge them later on. So, they get combined into one dataset so their names of provinces can be changed in one go, which is done below this.

```
name_change[1,1] <- "Groningen"
name_change[2,1] <- "Fryslân"
name_change[3,1] <- "Drenthe"
name_change[4,1] <- "Overijssel"
name_change[5,1] <- "Flevoland"
name_change[6,1] <- "Gelderland"
name_change[7,1] <- "Utrecht"
name_change[8,1] <- "Noord-Holland"
name_change[9,1] <- "Zuid-Holland"
name_change[10,1] <- "Zeeland"
name_change[11,1] <- "Noord-Brabant"
name_change[12,1] <- "Limburg"
```

```
dataset_map <- left_join(data_overweight_provence, name_change)
```

```
## Joining with ‘by = join_by(Regio)‘
```

Now these dataset can be merged into one complete dataset, the second final dataset.

```
proc_ver_overgewicht = ((overgewicht_2025 - overgewicht_1981) / overgewicht_1981) * 100
proc_ver_overgewicht[[1]]
```

```
## [1] 63.50365
```

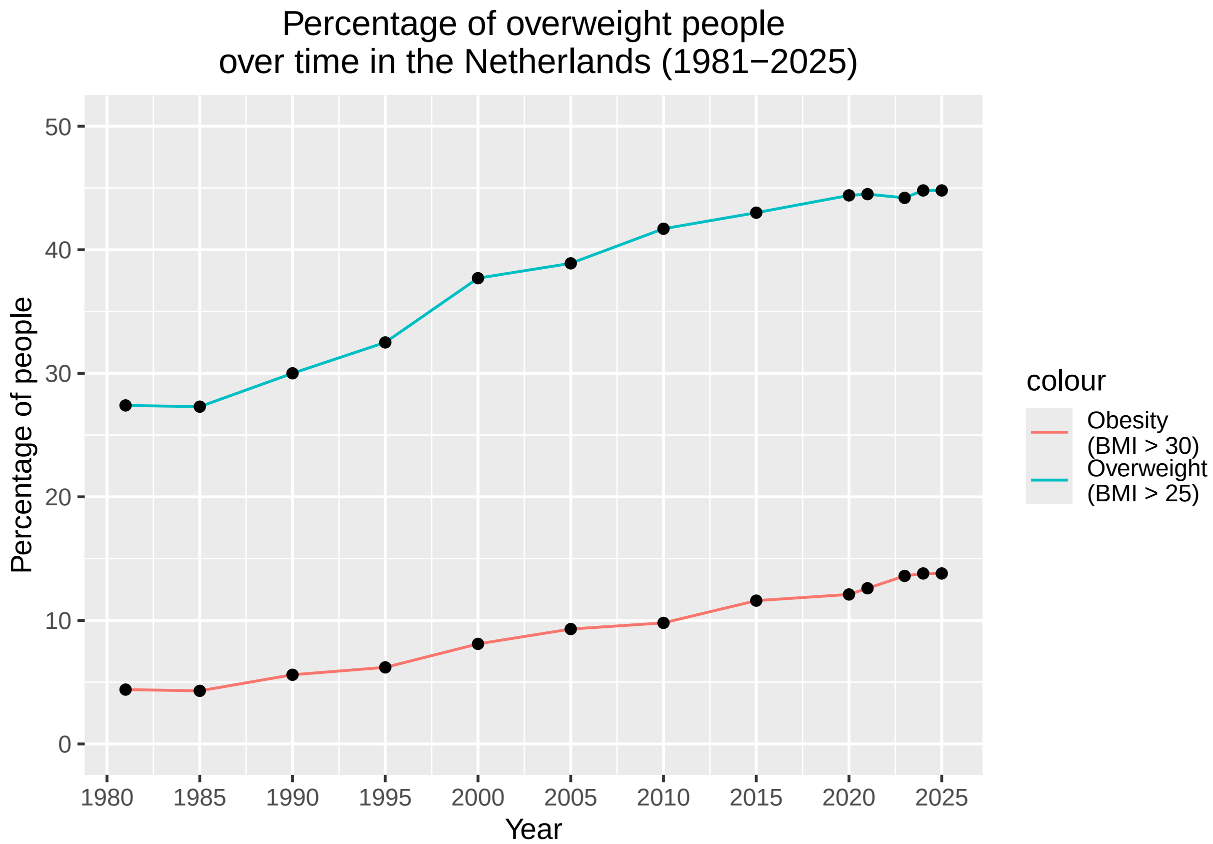
To show the importance of this research a new variable is created, the percentage change of overweight people in the Netherlands is calculated over 44 years, which is 63.5 percent. This is obviously a huge increase which need to be addressed and more attention given to.

3.3 Overweight and obesity over time

```
perc_over_time_plot = ggplot() +
  geom_line(aes(x = perc_overgewicht[[1]],
                y = perc_overgewicht[[2]],
                color = "Overweight \n(BMI > 25)")) +
  geom_line(aes(x = perc_overgewicht[[1]],
                y = perc_overgewicht[[3]],
                color = "Obesity  \n(BMI > 30)")) +
  geom_point(aes(x = perc_overgewicht[[1]],
                 y = perc_overgewicht[[2]])) +
  geom_point(aes(x = perc_overgewicht[[1]],
                 y = perc_overgewicht[[3]])) +
  scale_x_continuous(breaks = seq(1980, 2025, by = 5)) +
  ylim(0,50) +
  labs(
    title = "Percentage of overweight people\n over time in the Netherlands (1981-2025)",
    x = "Year",
    y = "Percentage of people") +
    theme(plot.title = element_text(hjust = 0.5))

ggsave("Graphs/Percentage_overgewicht_over_tijd.png",
       perc_over_time_plot, width = 9, height = 6)

perc_over_time_plot
```



A plot gets created with ggplot, which gets saved with ggsave. geom_line and geom_point get used to create a clear plot, with scale_x_continuous the x axis values get determined.

As shown in the graph, the percentage of overweight/obese people has increased at a relatively constant rate over time, with the exception of a larger increase between 1995 and 2000.

3.4 Maps of different provinces in the Netherlands for overweight, education and income

```
provincies_kaart <- st_read(paste0(
  "https://service.pdok.nl/cbs/gebiedsindelingen/2024/wfs/v1_0?",
  "request=GetFeature&service=WFS&version=2.0.0",
  "&typeName=provincie_gegeneraliseerd&outputFormat=application/json"
))
```

```
## Reading layer ‘provincie_gegeneraliseerd’ from data source
##   ‘https://service.pdok.nl/cbs/gebiedsindelingen/2024/wfs/v1_0?request=GetFeature&service=WFS&versio
##   using driver ‘GeoJSON’
## Simple feature collection with 12 features and 5 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 13565.4 ymin: 306846.2 xmax: 278026.1 ymax: 619231.6
## Projected CRS: Amersfoort / RD New
```

```r
provincies_kaart$statnaam
```

```
##  [1] "Groningen"     "Fryslân"       "Drenthe"       "Overijssel"
##  [5] "Flevoland"     "Gelderland"    "Utrecht"       "Noord-Holland"
##  [9] "Zuid-Holland"  "Zeeland"       "Noord-Brabant" "Limburg"
```

```r
nl_map <- provincies_kaart |> left_join(dataset_map, by = c("statnaam" = "Regio"))
```

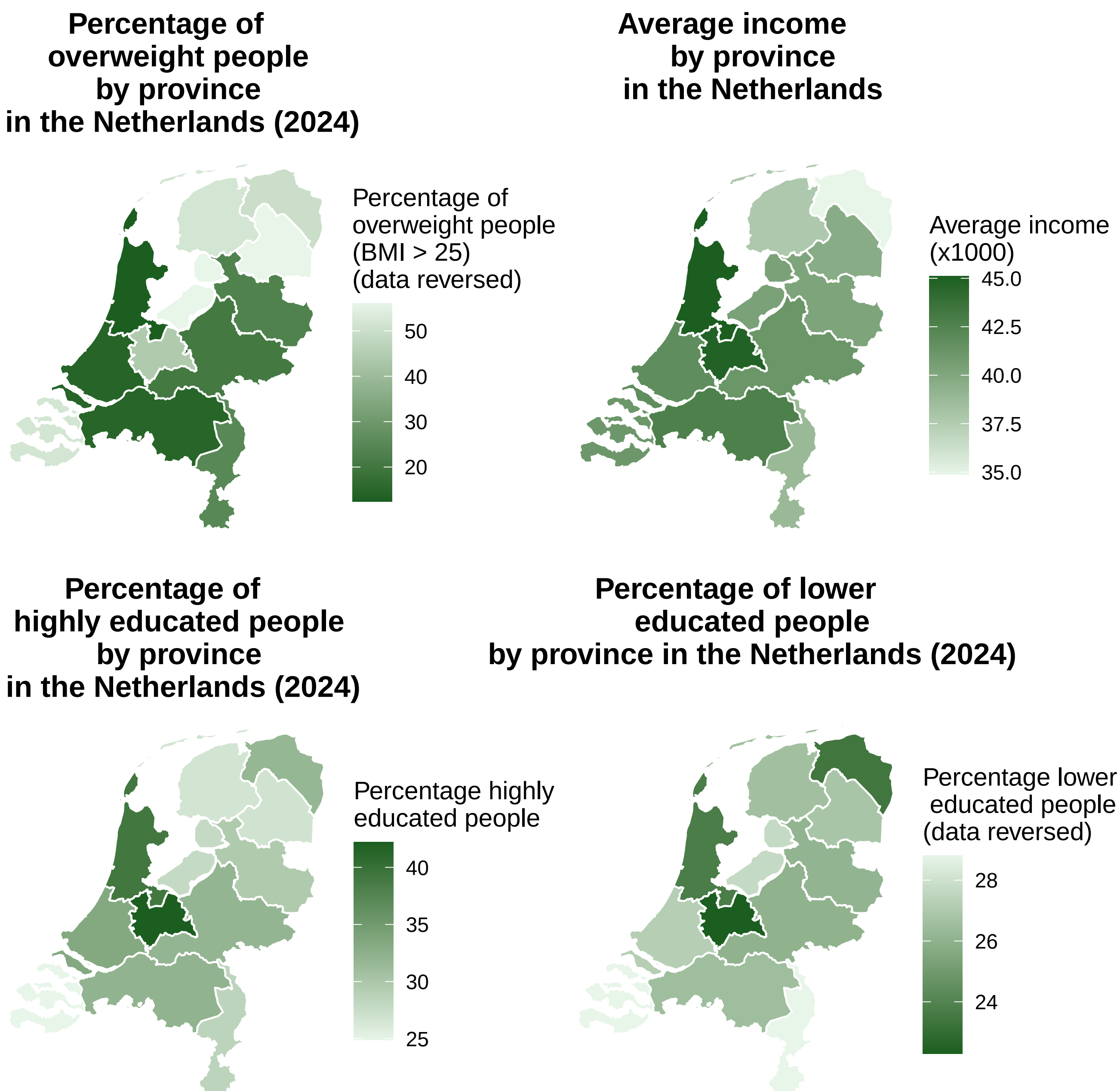
One dataset gets merged with the provinces map of the Netherlands, to make 4 different maps to compare them.

```r
combined_plot <-
  (ggplot(nl_map) +
  geom_sf(aes(fill = Overgewicht ), color = "white", linewidth = 0.4) +
  scale_fill_gradient(name = "Percentage of \noverweight people \n(BMI > 25)\n(data reversed)",
                      high = "#e8f5e9", low = "#1b5e20") +
  labs(
    title = "Percentage of
    overweight people
    by province
    in the Netherlands (2024)") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
  |
  ggplot(nl_map) +
  geom_sf(aes(fill =  Gemiddeld_inkomen.x.1000.), color = "white", linewidth = 0.4) +
  scale_fill_gradient(name = "Average income \n(x1000)",
                      low = "#e8f5e9", high = "#1b5e20") +
  labs(
    title   = "Average income
    by province
    in the Netherlands"  ) +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))) /

  (ggplot(nl_map) +
  geom_sf(aes(fill = Percentage_hoogopgeleide ), color = "white", linewidth = 0.4) +
  scale_fill_gradient(name = "Percentage highly \neducated people",
                      low = "#e8f5e9", high = "#1b5e20") +
  labs(
    title = "Percentage of
    highly educated people
    by province
    in the Netherlands (2024)") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
 |
   ggplot(nl_map) +
  geom_sf(aes(fill = Percentage_laagopgeleide ), color = "white", linewidth = 0.4) +
  scale_fill_gradient(name = "Percentage lower\n educated people \n(data reversed)",
                      high = "#e8f5e9", low = "#1b5e20") +
  labs(
```

```
    title = "Percentage of lower
    educated people
    by province in the Netherlands (2024)") +
  theme_void() +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))
   )

combined_plot
```



The dataset for the maps gets used for four different plots. These get combined into one big plot.

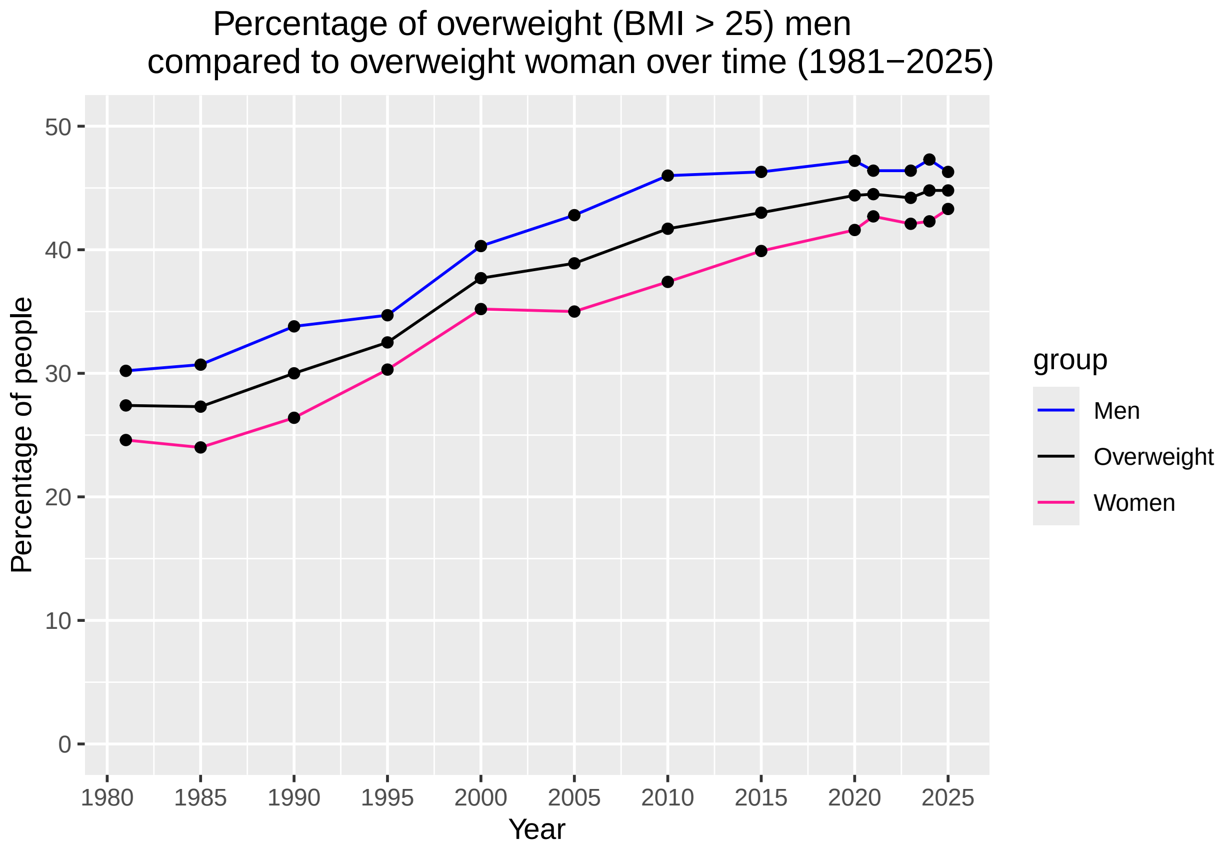
In the top left and down right, the data is reversed. This is done because this makes comparing the different maps easier and clearer.

As shown in the map representing the perecentage of overweight and obese individuals, the provinces Utrecht, Noord-Holland, and Groningen appear to have the lowest prevalence rates. In contrast, Flevoland, Limburg and Drenthe have the highest percentage of overweight and obese residents. Furthermore a possible correlation can be found between overweight and obesity and educational level. The provinces with the lowest prevalence rates also have the highest level of education, whereas Flevoland and Drenthe, which have a relatively lower education level, show a higher percentage of overweight and obese residents. The level of income and overweight appear to have a weaker relationship.

3.5 Overweight for men and women

```
plot_men_vs_women <- ggplot() +
  geom_line(aes(x = perc_overgewicht[[1]],
              y = perc_overgewicht[[2]],
              color = "Overweight")) +
  geom_line(aes(x = perc_overgewicht[[1]],
              y = perc_overgewicht[[4]],
              color = "Men")) +
  geom_line(aes(x = perc_overgewicht[[1]],
              y = perc_overgewicht[[5]],
              color = "Women")) +
  geom_point(aes(x = perc_overgewicht[[1]],
              y = perc_overgewicht[[2]])) +
  geom_point(aes(x = perc_overgewicht[[1]],
               y = perc_overgewicht[[4]])) +
  geom_point(aes(x= perc_overgewicht[[1]],
             y = perc_overgewicht[[5]]))  +
  scale_color_manual( values = c("Overweight" = "black",
                                "Men" = "blue",
                                "Women" = "deeppink")) +
  scale_x_continuous(breaks = seq(1980, 2025, by = 5)) +
  ylim(0,50) +
  labs(title = "Percentage of overweight (BMI > 25) men
       compared to overweight woman over time (1981-2025)",
       x = "Year",
       y = "Percentage of people",
       color = "group") +
    theme(plot.title = element_text(hjust = 0.5))

ggsave("Graphs/plot_men_vs_women.png", plot_men_vs_women, width = 9, height = 6)
plot_men_vs_women
```



The plot gets created with ggplot and is saved with ggsave. geom_line and geom_point are used to create a clear plot, with scale_x_continuous the x axis values get determined.scale_color_manual gives the lines the correct colors. labs is used to create the labels/text in the graph.

Over the 44-year period shown, the percentage of overweight adults has always been higher among men than among women. Although this gap has shrunk in recent years, men still show a higher rates of overweight. furthermore, the percentage of overweight men and the percentage of overweight women have generally moved together over time.

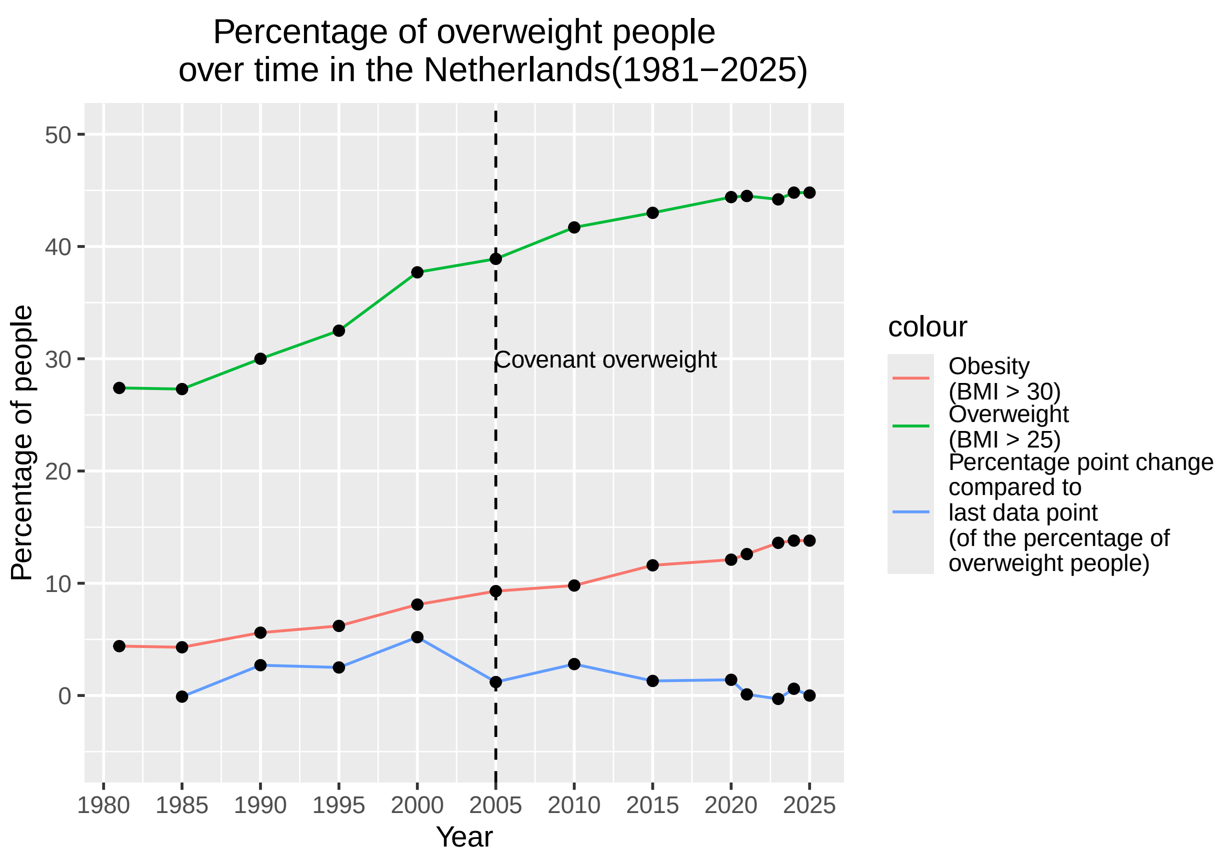
3.6 Dutch covenant overweight

in 2005, the Dutch covenant overweight was introduced and signed by several organizations with the goal of preventing overweight and obesity. Here it is tested if the percentage of overweight and obese people has reduced since its introduction.

```
plot_event <- ggplot() +
  geom_line(aes(x = perc_overgewicht[[1]],
                y = perc_overgewicht[[2]],
                color = "Overweight \n(BMI > 25)")) +
  geom_line(aes(x = perc_overgewicht[[1]],
                y = perc_overgewicht[[3]],
                color = "Obesity \n(BMI > 30)")) +
  geom_line(aes(x = perc_overgewicht[[1]],
                y = perc_overgewicht[[6]],
              color = "Percentage point change\ncompared to\nlast data point\n(of the percentage of \nov
              na.rm = TRUE) +
```

```
    geom_point(aes(x = perc_overgewicht[[1]],
                   y = perc_overgewicht[[2]])) +
  geom_point(aes(x = perc_overgewicht[[1]],
                   y = perc_overgewicht[[3]])) +
  geom_point(aes(x = perc_overgewicht[[1]],
                  y = perc_overgewicht[[6]]),
                na.rm = TRUE) +
  geom_vline(xintercept = 2005, linetype = "dashed")+
  annotate("text", x = 2012, y = 30, size = 3, label = "Covenant overweight") +
  scale_x_continuous(breaks = seq(1980, 2025, by = 5)) +
  ylim(-5,50) +
  labs(title = "Percentage of overweight people
      over time in the Netherlands(1981-2025)",
       x = "Year",
       y = "Percentage of people") +
    theme(plot.title = element_text(hjust = 0.5))

ggsave("Graphs/®plot_event.png", plot_event, width = 9, height = 6)
plot_event
```



The plot gets created with ggplot and is saved with ggsave. geom_line and geom_point are used to create a clear plot, with scale_x_continuous the x axis values get determined.geom_vline is for creating a vertical line at the year of the event and annotate is used for adding the event name next to the vertical line. Labs is used to create the labels/text in the graph.

As shown in the temporal variation, the percentage of overweight/obese people has increased at a relatively constant rate over time. This rate seems a to have slowed down a little from 2005 onwards in absolute terms, but this could be due to multiple reasons. Consequently, it is hard to determine a clear change following the introduction of the covenant on overweight and obesity, but an argument could be made that it did make a change.

4 - Discussion

The results of this study show that overweight and obesity have become increasingly important public health issues in the Netherlands. Between 1981 and 2025, the percentage of overweight individuals increased by approximately 63.5%, highlighting a strong upward trend over time.

Our regional analysis suggests a relationship between socioeconomic status and overweight prevalence. Provinces with a higher share of highly educated residents, such as Utrecht and Noord-Holland, generally have lower rates of overweight, while provinces with lower education levels, such as Drenthe and Flevoland, show higher rates. In contrast, the relationship between income and overweight appears less clear, suggesting that education may be more strongly associated with overweight than income.

We also found that men consistently have higher rates of overweight than women, although both groups follow a similar increasing trend over time. Furthermore, the introduction of the Dutch Covenant on Overweight in 2005 does not appear to have caused a clear reduction in overweight prevalence, although the growth rate may have slowed slightly.

A limitation of this study is that it relies mainly on descriptive analysis and visual comparisons, meaning that causal relationships cannot be established. Nevertheless, the findings indicate that overweight is linked to broader socioeconomic inequalities and should be viewed as more than just an individual lifestyle issue.

5 Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here: https://github.com/davidhers/PfE_2.6_2026.git

5.2 Reference list

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